2nd Midterm Notes

IT 110
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A number in exponential notation is represented as:

$$\underbrace{MMMM}_{\text{Mantissa}} \times \underbrace{BBBB}_{\text{Base}}{}^{\overbrace{EEEEE}^{\text{Exponent}}}$$

Example: $2 \times 10^2 =$

$= 2 \times 100 = 200$

Mantissa, base and exponent need not to be integers.

1) Example 1: Convert 111.625 → to IEEE 754

Step 1: Integer part: →

$111 = 64 + x \Rightarrow x = 111 - 64 = 47$

$47 = 32 + y \Rightarrow y = 47 - 32 = 15$

$15 = 8 + 7 = 8 + 4 + 2 + 1$

So $111 = 64 + 32 + 8 + 4 + 2 + 1$

$+64_{DEC} = 2^6_{DEC} = 1000\ 000_B$

$+32_{DEC} = 2^5_{DEC} = 0100\ 000_B$ $+$

$+8_{DEC} = 2^3_{DEC} = 0001\ 000_B$

$+4_{DEC} = 2^2_{DEC} = 0000100_B$

$+2_{DEC} = 2^1_{DEC} = 0000010_B$

$+1_{DEC} = 2^0_{DEC} = \underline{0000001_B}$

1101111

2) Fractional Part

x	$*2^1$	$x - \lfloor x \rfloor$
0.625_D	1.25_D	0.25_D
0.25_D	0.50_D	0.50_D
0.50_D	1.00_D	0.00_D

$0.625_{DEC} = 101_{BIN}$

3) Add integer binary and fractional binary:

$N = Int_B + Frac_B = 1101111.101$

4) Apply a right bit shift of 6 places:

$= 1101111101 \times 2^6$. Remember single precision IEEE754 floating point numbers are 1 sign bit 8 bits Exponent with bias 127 (excess 127) and 23 bits mantissa so;

$= 1101111.101 \times 2^{133}$

So let's again convert 133_{DEC} to EEEEEEEE$_{BIN}$

Trick:

Get closest power of 2 to 128

this is: 2^7 or ~~10000000~~

$133 = 128 + 5 = 128 + 4 + 1$

$$
\begin{array}{rr}
128_{DEC} = & 1000\ 0000 \\
4_{DEC} = & 0000\ 0100 \ + \\
+ \quad 1_{DEC} = & 0000\ 0001 \\
\hline
 & 1000\ 0101
\end{array}
$$

So our number is

$\underbrace{0}_{(+)}\ \underbrace{10000101}_{\text{EXPONENT}}\ \underbrace{10111110100000000000000}_{\text{MANTISSA w/ 14 Trailing 0's.}}$

(B) Convert 65.2

1)
$$
\begin{array}{rl}
32_{DEC} = & 2^5_D = 0100000_{BIN} \\
+\ 32_{DEC} = & 2^5_D = 0100000_{BIN} \\
01_{DEC} = & 2^0_D = 0000001 \\
\hline
65 & 1000001
\end{array}
$$

2)

x	$*2^1$	$x - \lfloor x \rfloor$
0.2	0.4	0.4
0.4	0.8	0.8
0.8	1.6	0.6
0.6	1.2	0.2
⋮	⋮	⋮

(1 period is enough here)

So $0.2_{DEC} = 0.\overline{0011}_{BIN}$

This becomes an infinite periodic number!

3) Add integer and fractional parts:

$$N = 1000001_B + 0.\overline{0011}_B = 100\,0001.0011_B$$

4) Right shift until most significant bit is the only integer:

$$N >> 6_{DEC} = 1.0000010011_B * 1000000_B$$

So $6 + \underbrace{127}_{\text{Bias/Excess}} = 133$

5) from first exercise

$$133_{DEC} = 10000101_{BIN}$$

6) $\underset{(+)}{0}\;\underbrace{10000101}_{\text{EXPONENT}}\;\underbrace{000001\overset{1}{\overline{0011}}\overset{2}{\overline{0011}}\overset{3}{\overline{0011}}\overset{4}{\overline{0011}}0}_{\text{MANTISSA}}$

MANTISSA → 23bit = 6 bits + 4 × 4-bit periods + 1 bit

(c)

$(-)\underbrace{770}.17_{DEC} = \text{IEEE } 754\ ?$

Closest preceding power of 2 (2^9)

1) $770 = 512 + x \Rightarrow x = 770 - 512 =$

$$\begin{array}{r} 770 \\ -\ 512 \\ \hline 258 \end{array} \qquad \begin{array}{r} 258 \\ -\ 256 \\ \hline 002 \end{array} \to 2^1$$

~ Closest preceding pow. of 2 (2^8)

9 0's

$$\begin{array}{rcl} \text{So } 512_{DEC} = 2^9_{DEC} = & 1000\ 000\ 000 \\ +\ 256_{DEC} = 2^8_{DEC} = & 0100\ 000\ 000 \\ 002_{DEC} = 2^1_{DEC} = & 0000\ 000\ 010 \\ \hline & 1100\ 000\ 010 \end{array}$$

2)

$x = 0.17$	2^1	$x - \lfloor x \rfloor$
0.17	0.34	0.34
0.34	0.68	0.68
0.68	1.36	0.36
0.36	0.72	0.72
0.72	1.44	0.44
0.44	0.88	0.88
0.88	1.76	0.76
0.76	1.52	0.52
0.52	1.04	0.04
0.04	0.08	0.08

So: $x \approx 0.001010111$

At the Least truncation error, STOP!

So: $\approx$ means approximately

$$1100000000 1.001010111 \approx 770.17$$

Now apply a right bit shift of 8 to get the number in IEEE 754 format:

$$\underbrace{11000000001}_{9 \text{ bits}}.001010111 = \ldots$$

$$\ldots = 1.1000000001000101011 1 \times 1\underbrace{0000}\underbrace{0000}0$$

2^9 in excess 127 is: $2^{E-127=9}$

$\Rightarrow E = 127 + 9 = 136$

$136_{DEC} = 128 + X \Rightarrow X = 136 - 128 = 8$

$$\begin{array}{rl} 128_D = & 0100\,0\,0000_{BIN} \\ +\ 008_{DEC} = & 0000\,0\,1000_{BIN} \\ \hline +\ 136_{DEC} = & 1000\,1000 \end{array}$$

$$\Rightarrow N = \underset{(-)}{1}\ \underbrace{\overset{8 \text{ bits}}{10001000}}_{\text{EXPONENT}}\ \underbrace{100000000100010101111}_{18 \text{ bits - Significand}}\ \underbrace{00000}_{\text{five } 0\text{'s}}$$

$$
\begin{array}{r}
770.17 \\
-\ 065.20 \\
\hline
?
\end{array}
$$

1. First you must make exponents equal since for n different from m:

$$u \cdot 2^{n} + v \cdot 2^{m} = u' \cdot 2^{q} + v' \cdot 2^{q}$$

$$(u' + v') \cdot 2^{q}, \text{ where } q = n \cdot m$$

and $u' = u \cdot 2^{-m}$, $v' = v \cdot 2^{-n}$

So u' is the result of a right shift by m bits of u, likewise v' is the result of a right shift of v by n bits.

$770.17_{DEC} =$

$= 0 \mid \underbrace{10001000}_{\text{EXPONENT}} \mid 1000000\ 10001010111\ 00000$ IEEE 754

$-65.20_{DEC} =$

$= 1 \mid \underbrace{10000101}_{\text{EXPONENT}} \mid 00000\ \overline{10011}^{1}\ \overline{0011}^{2}\ \overline{0011}^{3}\ \overline{00110}^{4}$ periods IEEE 754

+11 ~ 3 right shifts (See below (A))

Tip: note 65.02's exponent is smaller than the one of 770.17

65.02's exponent subtracted from that of 770.17 is:

(A)

$$
\begin{array}{r}
10001000 \\
-\ 10000101 \\
\hline
10001000 \\
+\ 01111011 \\
\hline
1\,00000011
\end{array}
$$

01111011 → 2's complement

00000011 → 3 right shifts

1 → Ignore carry (no underflow)

For now we allow -65.20 to become "subnormal" meaning we will bring in the most significant digit into the Mantissa and the Mantissa will have a 0 instead

	EXPONENT								MANTISSA																								
	0	1	0	0	0	1	0	0	0	1	0	0	0	0	0	0	1	0	0	0	1	0	1	0	1	1	1	0	0	0	0	1
(−)	1	1	0	0	0	1	0	0	0	0	0	1	0	0	0	0	0	1	0	0	1	1	0	0	1	1	0	0	1	1	0	1
2's	0	1	0	0	0	1	0	0	0	1	0	0	0	0	0	0	1	0	0	0	1	0	1	0	1	1	1	0	0	0	0	1
+	0	1	0	0	0	1	0	0	0	1	1	0	1	1	1	1	1	0	1	1	0	0	1	1	0	0	1	1	0	0	1	1
	0	1	0	0	0	1	0	0	0	0 1	1	0	0	0	0	0	0	1	1	1	1	1	0	0	0	0	1	0	1	0	0	0

The carry overflows so we take into account that, but we ignore it because we know that 65.02 is much smaller than 770.17 and our 23 bit mantissa space, i.e. 65 has extra leading zeros

Result is 704.97 ≈

0 1000 1000 01100000011111000010100

Exercise Verify the last result converting 704.97 to IEEE754

LMC

Program to add

00	901	IN	(Basket) Slowest operation
01	3xx	STO	Store at mailbox 99
02	901	IN	
03	199	ADD	# Stored at 99 with current
04	902	OUT	
05	000	HLT	
99		DAT	

Visit the address in the MOODLE INFO.

BRP - Branch at Zero or Positive

This allows to implement reps of instructions repetitively

For instance *TO CAL means "to calculator"

MNEMONIC	MLBOX/OP-	OP	WHERE	RESULT
Load LDA(5)	000	5	12 (001)	001 → TO CALC
add ADD (1)	001	1	15 (390)	+ = 391 → TO CAL
Store STO(3)	002	3	04	391
Load LDA(5)	003	5	12 (001)	001
Halt HLT (0)	000	—	—	000
store STO (3)	004	3	91 (001)	001
Load LDA (5)	005	5	12	001
Add ADD (1)	006	1	13 =	+ +001
Store STO (3)	007	3	12	002
Load LDA(5)	008	5	14	005
Subtract SUB(2)	009	2	12 (002)	003
BRANCH POSITIVE BRP(8)	010	8	00	